

About

This project focuses on portfolio performance and attribution analysis, presented through an interactive Power BI dashboard designed to evaluate and visualize investment performance. The dataset includes benchmark index data and portfolio data derived from an investment fund's performance.

Key Components

- Data Processing: Data cleaning and transformation were conducted using Excel and Power Query.
- Calculations & Analysis: Implemented using DAX, including portfolio weight calculations, the Brinson attribution model for performance attribution, and key rate duration (KRD) analysis for fixed-income exposure.
- Dashboard Structure: The dashboard features five key pages:
 1. Summary – Provides an overview of portfolio performance.
 2. Equity Analysis – Performs attribution analysis for equity holdings.
 3. Fixed Income Analysis – Breaks down fixed-income asset performance by sector and risk factors.
 4. Risk Analysis – Assesses portfolio risks using metrics such as volatility, tracking error, and value at risk (VaR).
 5. Scenario Analysis – Examines portfolio performance under various economic conditions.

Limitations & Future Improvements

- Data Constraints: The dataset spans only nine months with monthly data points, limiting the ability to calculate certain risk metrics (e.g., volatility, VaR, drawdown analysis) with historical accuracy.
- Scope: The dataset currently includes only Canadian equity and fixed-income assets. Future improvements include expanding the dataset to cover global markets, additional asset classes, and a longer time horizon.

Portfolio Summary 2024

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep
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Select a Date, Return Type, and Securities Type to view performance

MoM

YTD

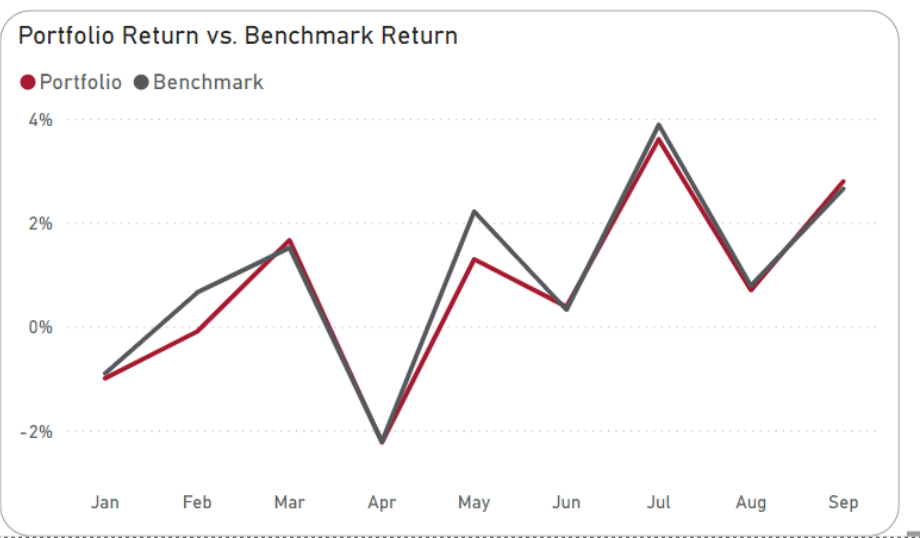
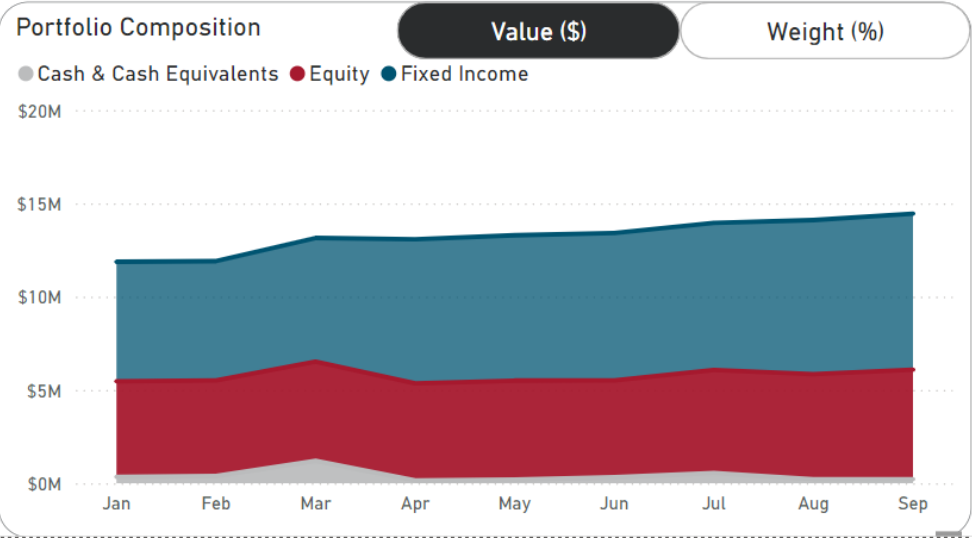
EQUITY

FIXED INCOME SECURITIES



Portfolio Value (Previous Month) \$13,411,668		Portfolio Value (Current) \$13,960,231		Portfolio Value Change \$548,563	
Portfolio Return 3.60%		Benchmark Return 3.88%		Active Return -0.28%	
Portfolio Volatility 1.95%	Tracking Error 0.45%		Information Ratio -0.63		# of Trades 2

Note: Risk metrics are calculated using month-over-month (MoM) data, based on the most recent 6 months, and are not annualized.



Equity

Fixed Income

2024

Jan

Feb

Mar

Apr

May

Jun

Jul

Aug

Sep

MoM Portfolio...

6.27%

MoM Benchmark...

6.38%

MoM Active Return

-0.11%

YTD Portfolio...

9.45%

YTD Benchmark...

10.64%

YTD Active Return

-1.20%

Tracking Error

0.69%

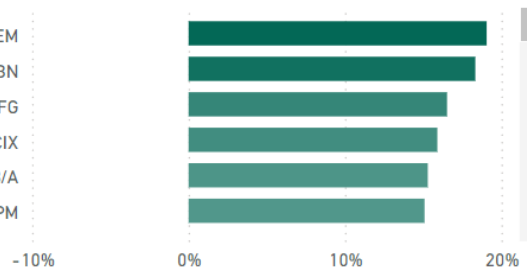
of Holdings

40

Sector/Security Return (%)

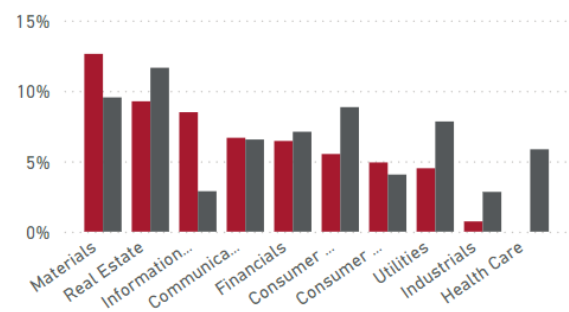
Dill up and down to view Return by Sector and Security

-6.47% 6.29% 19.05%

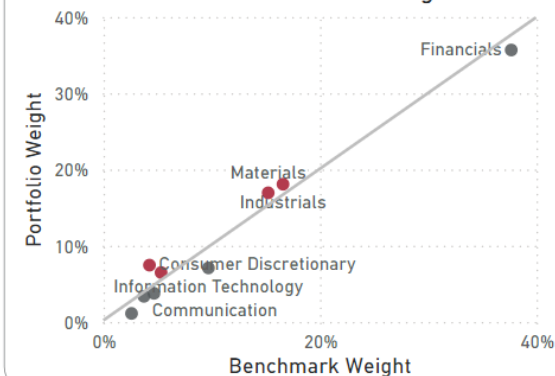
AEM
BN
WFG
CIX
GIB/A
WPM

Portfolio Return vs. Benchmark Return by Sector

● Portfolio ● Benchmark

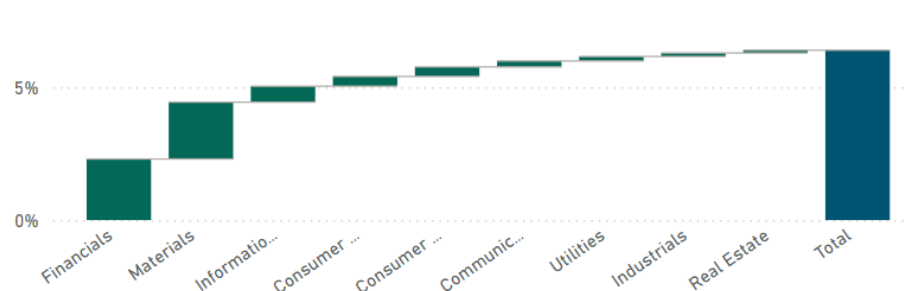


Portfolio vs. Benchmark Sector Weights



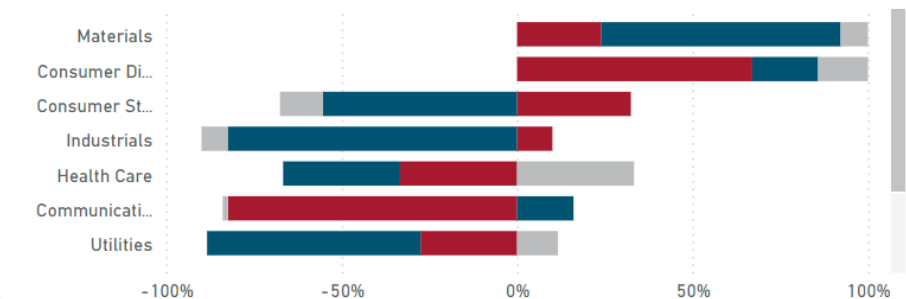
Return Contribution by Sector

● Increase ● Decrease ● Total



Active Return Attribution

● Allocation ● Selection ● Interaction



Equity

Fixed Income

All



Jan

Feb

Mar

Apr

May

Jun

Jul

Aug

Sep



MoM Portfolio Return

1.36%

MoM Benchmark Return

1.09%

MoM Active Return

0.27%

Portfolio Duration

6.89

Benchmark Duration

7.23

Duration Position

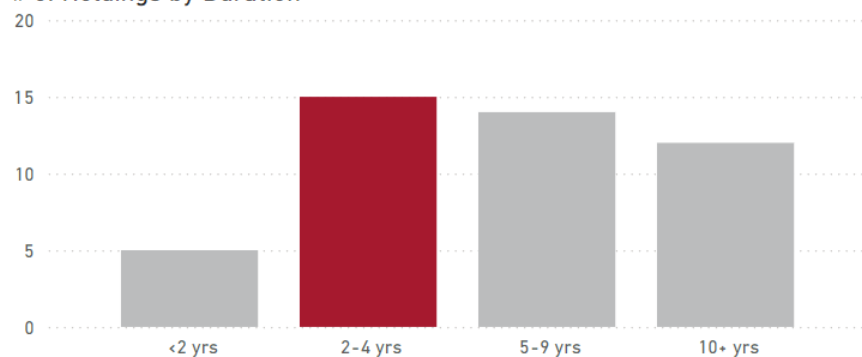
-0.33



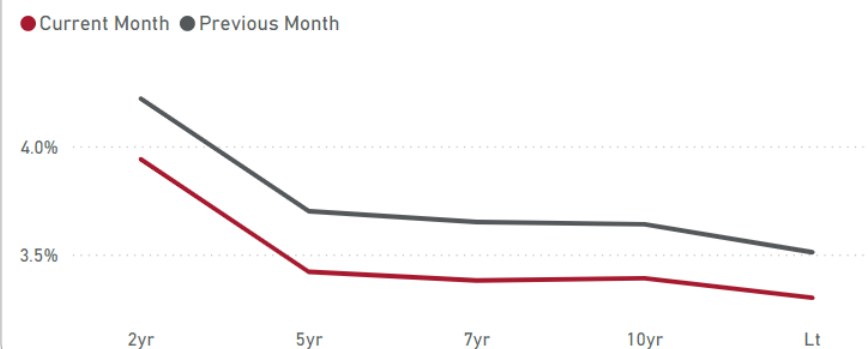
Sector

Factor

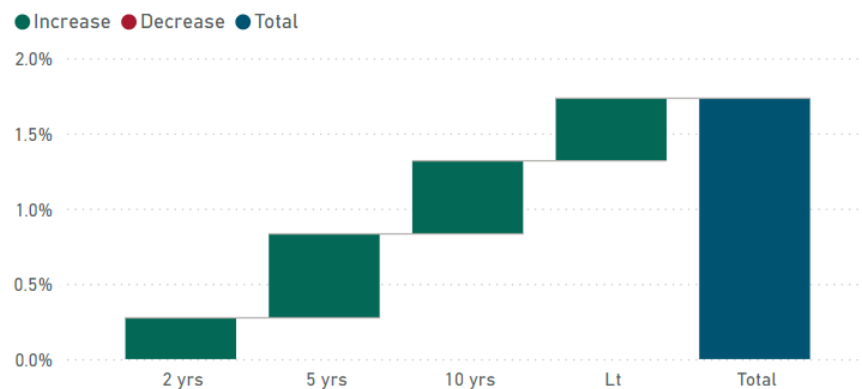
of Holdings by Duration



Term Structure of Interest Rate

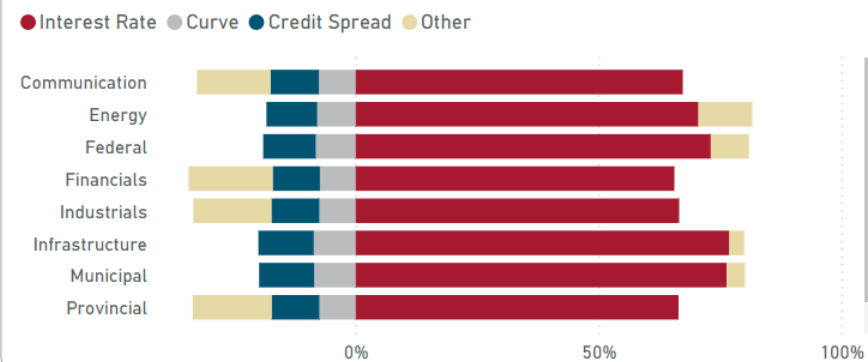


Key Rate Duration Analysis



Return Attribution by Factor

Dill up and down to view attribution by Sector Group and Sector



Risk Analysis

2024

Jan

Feb

Mar

Apr

May

Jun

Jul

Aug

Sep

EQUITY

FIXED INCOME SECURITIES

Portfolio Return

3.60%

Benchmark

3.88%

Portfolio Volatility

1.95%

Benchmark

1.78%

Portfolio Sharpe Ratio

-0.09

Benchmark

0.06

Active Return

-0.28%

Tracking Error

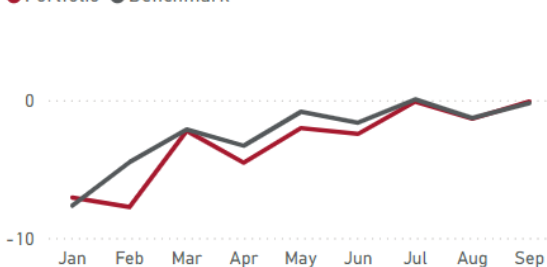
0.45%

Information Ratio

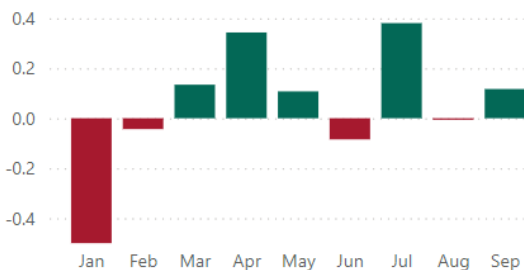
-0.63

Portfolio vs. Benchmark Sharpe Ratio

● Portfolio ● Benchmark

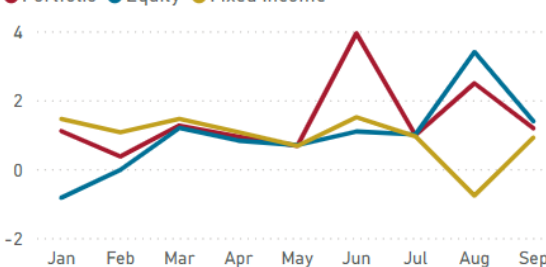


Correlation Between Equity & Fixed Income

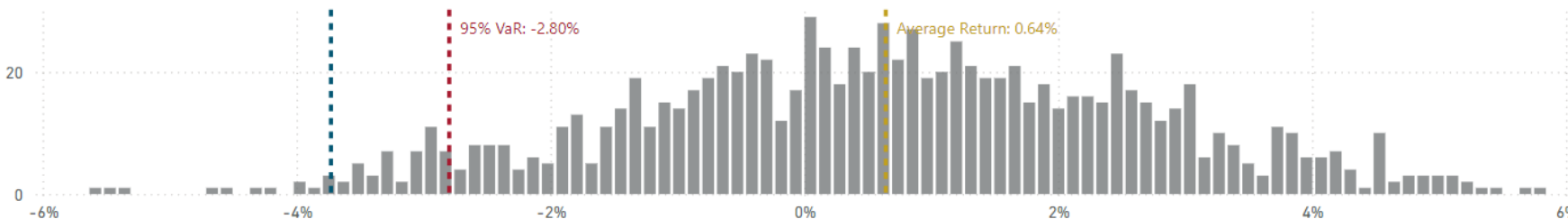


Beta Over Time

● Portfolio ● Equity ● Fixed Income



Portfolio VaR



Scenario Analysis

Results



Scenario	Equity Portfolio	Fixed Income Portfolio	Total Impact
Base Case	2.95%	0.10%	1.25%
Inflation	5.31%	-3.61%	0.11%
Recession	-4.36%	2.24%	-0.50%
Stagflation	0.66%	-0.88%	-0.23%

Assumptions

Scenario	2yr Rate	5yr Rate	10yr Rate	30yr Rate	Growth Stock	Cyclical Stock	Defensive Stock
Base Case	-0.10%	-0.05%	0.00%	0.05%	5.00%	3.00%	1.00%
Inflation	0.70%	0.60%	0.50%	0.40%	12.00%	5.00%	3.00%
Recession	-0.50%	-0.40%	-0.30%	-0.20%	-10.00%	-5.00%	5.00%
Stagflation	0.30%	0.20%	0.10%	0.00%	-2.00%	0.00%	8.00%